Reflective Writing Example: Survival Analysis

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> **Teaching note.** This is a fictional example designed to demonstrate how description, evidence and critical reflection can be integrated into one coherent response. The data and numerical results are illustrative only.

Reflective Writing Example

During my survival analysis coursework, I investigated the time until patients experienced a specified clinical event. I began by exploring the dataset and identifying the follow-up time, event indicator and relevant explanatory variables. I noticed that several patients had not experienced the event by the end of the study and were therefore right-censored. This made me realise why standard methods for analysing continuous outcomes would not be appropriate, as censoring needs to be accounted for when analysing time-to-event data (Collett 2023). I therefore used the Kaplan–Meier estimator (Kaplan and Meier 1958) to estimate the survival function and compared the survival experience of two patient groups (see Figure 1 in Appendix A).

The Kaplan–Meier curves suggested that the two groups had different survival experiences, although the uncertainty increased later in the study as fewer patients remained at risk. I initially focused mainly on whether the log-rank test gave a statistically significant result. However, when interpreting the results, I realised that statistical significance alone did not tell me how large or practically important the difference was. This led me to fit a Cox proportional hazards model (Cox 1972) and report estimated hazard ratios together with 95% confidence intervals (see Table 1 in Appendix A). The R code used for the analysis is included in the same Quarto document so that the analysis can be reproduced.

Fitting the Cox model made me think more carefully about model assumptions, particularly the proportional hazards assumption (Collett 2023). At first, I viewed checking the proportional hazards assumption as simply another step that needed to be completed after fitting the model. However, when I examined the Schoenfeld residuals (Figure 2 in Appendix A), I began to understand that the diagnostic process could directly affect whether my interpretation of the hazard ratios was reasonable. I also recognised that successfully fitting a model does not necessarily mean that it provides an appropriate representation of the data.

If I were carrying out this analysis again, I would consider model assumptions and diagnostics much earlier rather than treating them as final checks. If there were evidence against proportional hazards, I would explore alternatives such as a stratified Cox model or time-varying effects and compare how these changed my conclusions. I would also place less emphasis on obtaining a significant p-value and more emphasis on effect sizes, uncertainty and whether the results were meaningful in the context of the research question. Overall, this analysis changed my understanding of survival analysis from a sequence of statistical techniques into a process of making and evaluating modelling decisions, where the assumptions, evidence and interpretation all need to support the conclusions I draw.

Appendix A: Supporting evidence

This appendix provides the evidence referred to in the reflective writing example. In a real submission, the figures, tables and code should come from the student's own work.

A1. Kaplan–Meier analysis

The figure above is evidence that a Kaplan–Meier analysis was carried out. In the main response, the student should still explain why the method was appropriate and reflect on what was learned from using it.

A2. Cox proportional hazards model

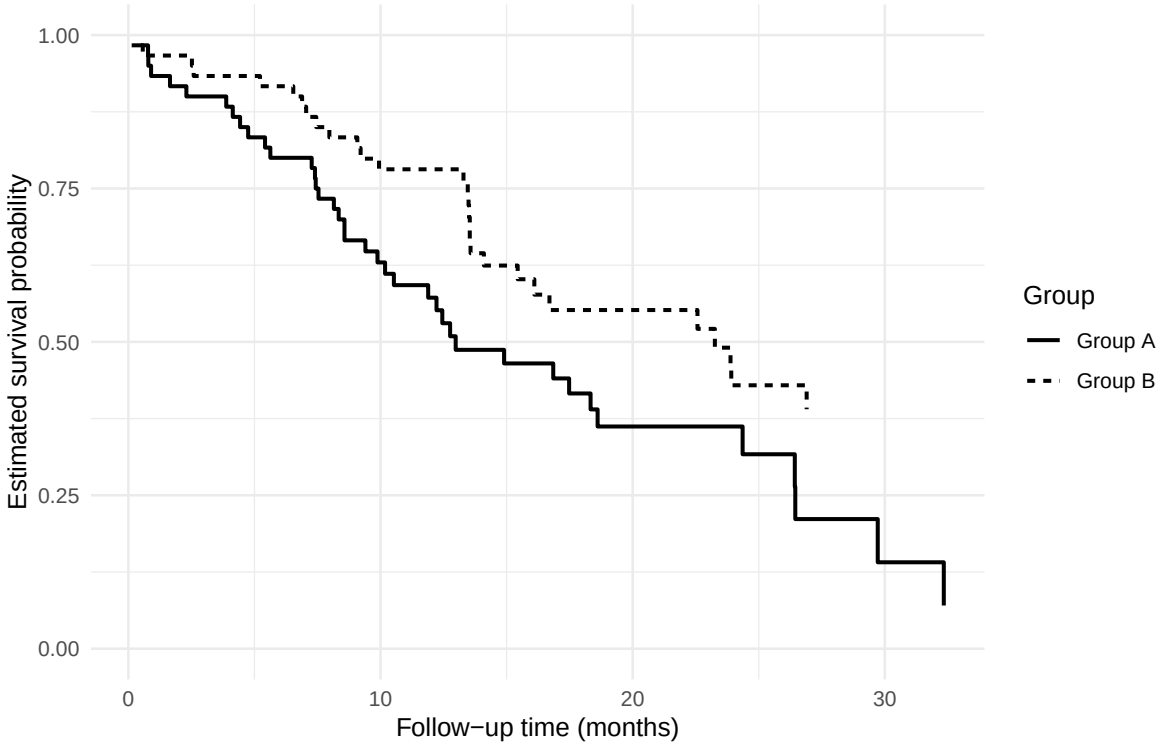

Figure 1: Kaplan–Meier estimates for the two illustrative patient groups.

Table 1: Illustrative Cox proportional hazards model results.

	Variable	Hazard ratio	95% CI	p-value
groupGroup B	Group B vs Group A	0.58	0.36–0.95	0.0315
I(age/10)	Age (per 10 years)	1.26	1.02–1.57	0.0343
severity	Baseline severity (per unit)	1.19	0.87–1.62	0.2787

The table provides evidence for the model-based statements made in the reflective writing. A student should avoid copying large quantities of software output when a small, clearly labelled table provides the relevant evidence more effectively.

A3. Model diagnostics

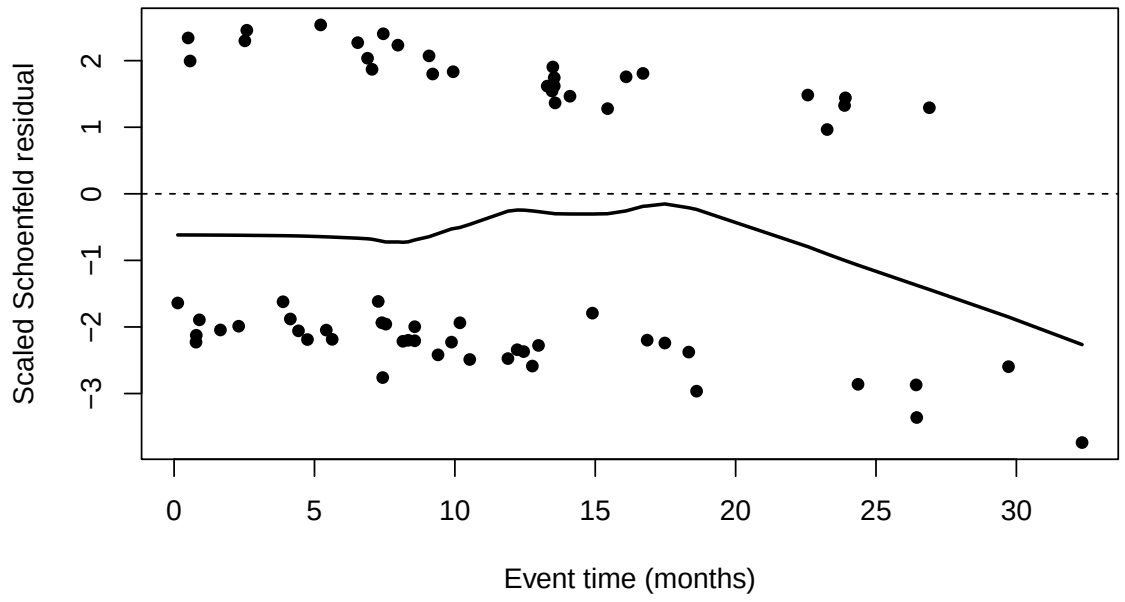

Figure 2: Scaled Schoenfeld residual diagnostic for the treatment-group effect.

This diagnostic is included because the main response specifically reflects on the proportional hazards assumption. Evidence in an appendix should be selected because it supports something discussed in the response, rather than being included simply because the software produced it.

A4. Reproducible code

The complete code required to reproduce Figure 1, Table 1 and Figure 2 is contained in this `.qmd` file. To make the code visible in the rendered document, change the YAML setting `execute: echo: false` to `execute: echo: true`.

A minimal version of the core analysis is:

```
library(survival)

dat <- read.csv("mock_survival_data.csv")

km_fit  <- survfit(Surv(time, event) ~ group, data = dat)
cox_fit <- coxph(Surv(time, event) ~ group + I(age/10) + severity, data = dat)

summary(cox_fit)
cox.zph(cox_fit)
```

References

Collett, David. 2023. *Modelling Survival Data in Medical Research.* 4th ed. Chapman; Hall/CRC.

Cox, D. R. 1972. "Regression Models and Life-Tables." *Journal of the Royal Statistical Society: Series B (Methodological)* 34 (2): 187–220.

Kaplan, Edward L., and Paul Meier. 1958. "Nonparametric Estimation from Incomplete Observations." *Journal of the American Statistical Association* 53 (282): 457–81.

How the evidence is referenced

Notice that the main reflective writing does not simply state that evidence exists in an appendix. It directs the reader to the relevant item at the point where the evidence matters: "see Figure 1 in Appendix A", "see Table 1 in Appendix A", and "Figure 2 in Appendix A".

The appendix supplies evidence of the student's own engagement; the main response provides the interpretation and critical reflection. Academic references, by contrast, support methodological statements and ideas drawn from books or papers.